# Saturated Superfluid Vacuum III
# Irreversible Time and Wake Entropy

Stig Norland

Draft

**Abstract**

This paper addresses only the origin of irreversible time in the Saturated Superfluid Vacuum (SSV) framework. We argue that time is not a background dimension through which reality moves, but the local rate of change of the vacuum order parameter $\Psi$. The past does not exist as a destination or region; it survives only as wake structure encoded in the present state of the medium. Events in $\Psi$ do not simply occur and vanish. They deposit distributed traces in the form of phonon phases, vortex distortions, Kelvin-wave content, density correlations, and altered future reconnection opportunities. We therefore propose that entropy in SSV should not be defined as a bare event count, but as the persistent wake-complexity of the present state.

On this view, irreversible time is not the flow of an external parameter, but the one-way accumulation and redistribution of wake structure in $\Psi$. Formal time reversal in equations is distinguished from physical reversal in the medium: the substitution $t \to -t$ defines a mathematical transformation on trajectories, but does not supply a physical operation that globally unwrites the distributed wake of an event. This dissolves the usual paradoxes of time travel by treating them as category errors arising from the reification of the past as a place.

## Contents

# 1 Introduction

The standard treatment of time in fundamental physics is powerful, but conceptually unstable. At the level of formal equations, time often appears as a reversible parameter. At the level of experience and thermodynamics, it appears as an irreversible order of change. This tension is usually managed rather than resolved.

The aim of this paper is narrow. We address only the origin of irreversible time in the SSV framework introduced in Paper I [1] and developed in Paper II [2]. The central claim is:

> Time is the local rate of change of $\Psi$, the past survives only as wake structure in the present state, and entropy is the persistent wake-complexity of that state.

This paper is therefore about ontology first and thermodynamic interpretation second. The goal is to state the time claim cleanly enough that the later papers can inherit it without having to carry the whole argument again.

## 1.1 Roadmap

Section 2 states the minimal SSV setting used in this paper. Section 3 states the ontological claim that time is local change rather than a background dimension, and draws the immediate consequence for time-travel paradoxes. Section 4 develops the idea that events write wake structure into the medium. Section 5 reframes entropy as wake-complexity and defines the five wake channels. Section 6 carries out the channel-wise reductions, working each channel to a recognizable entropy. Section 7 closes the remaining functional questions — subleading terms, the status of the regularization constants, the reduction of the five coarse-grainings to one renormalization flow, and the per-event accounting of reconnection. Section 8 distinguishes formal time reversal in equations from physical reversal in the medium. Section 9 closes with the remaining open technical questions.

# 2 Minimal SSV Setting

Only a small part of the SSV framework is needed here. The physical vacuum is represented by a macroscopic order parameter

$$\Psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)}\, e^{i\theta(\mathbf{x},t)}, \tag{1}$$

with density $\rho$, phase $\theta$, and superfluid velocity proportional to $\nabla\theta$ away from vortex cores. The exact field equation is not required for the present argument. What is required is that the state of the medium has real local structure: phase, density, vortex geometry, phonon content, correlations, and boundary data.

In this paper, an "event" means a localized update of that structure whose forward realization deposits wake content the medium cannot locally re-gather into the prior state. This criterion is deliberately structural rather than threshold-based. It admits any process whose forward and reverse boundary-data requirements differ  a reconnection, a phonon emission, a Kelvin-wave excitation, a vortex displacement that radiates  and excludes purely reversible relabellings and stationary zero-point fluctuations, whose statistics by construction contribute no net wake to $\mathcal{P}(t)$. Reconnection is the clearest illustrative case because it changes the connectivity of vortex filaments and converts localized order into distributed excitations, but it is not the only possible event, and it is not used here as a primitive counter for entropy.

The criterion is circular in form  wake is what events deposit, events are what deposit wake but the circularity is shared with any account of irreversibility, and it is closed empirically by the candidate functional in §5.3: a process counts as an event to the extent that it raises $W[\Psi]$ in any of its channels.

The present state of the medium will be denoted schematically by

$$\mathcal{P}(t) = \{\rho, \theta, \mathcal{V}, \mathcal{K}, \mathcal{C}, \ldots\}_t, \tag{2}$$

where $\mathcal{V}$ denotes vortex geometry, $\mathcal{K}$ Kelvin-wave content, and $\mathcal{C}$ correlation data. This notation is deliberately schematic. Its role is to keep the argument attached to present physical structure rather than to an abstract external time axis.

# 3 Time as Change, Not Dimension

In SSV, the physical vacuum is a real medium described by an order parameter $\Psi$. What physics ordinarily measures as time is not an independent dimension in which the medium is embedded, but the local rate at which the state of the medium changes. This is closer in spirit to the relational treatments of time advanced by Barbour [3] and Rovelli [4, 5] than to the block-universe reading common in spacetime physics.

A useful schematic statement is

$$d\tau \propto \mathcal{R}[\Psi]\, dt, \tag{3}$$

where $\mathcal{R}[\Psi]$ denotes the local update-rate of the medium. In later papers this quantity can be connected to gravitational and kinematic slowing. Here it is used only to express the ontological point: proper time is not a substance added to $\Psi$, but a measure of how much local physical change occurs in the medium.

This is not yet a final derivation, but a statement of ontology. The present is the current state of $\Psi$. The future does not yet exist. The past is not a region one could return to. It exists only insofar as earlier events have left structure in the present state of the medium.

## 3.1 Present-State Ontology and Causal Ordering

The strongest version of the claim has three parts:

1. only the present state of $\Psi$ physically exists,
2. the past is encoded in that state as record, wake, and correlation,
3. there is no second arena in which earlier states continue to exist as accessible places.

This does not deny ordering. Earlier and later remain meaningful because wake structure is asymmetric: a present configuration can contain records of a previous event (outgoing phases, displaced vortex geometry, thermalized phonons, modified reconnection opportunities), while the pre-event configuration does not contain the later wake as an already-existing region. Causal order is carried by structure in $\mathcal{P}(t)$, not by an external archive of completed moments.

A direct consequence is that time-travel paradoxes are not exotic physical possibilities but signs of a mistaken description. They rely on the assumption that the past remains available as a destination; in SSV there is no such region, and a journey "to" the past would require the current state of $\Psi$ to be both itself and an earlier state whose records it already contains. That is a contradiction in description, not a physical process to be analyzed.

# 4 Wake Production in $\Psi$

An event in $\Psi$ does not simply happen and disappear. It perturbs the medium and leaves a wake. Depending on scale and context, that wake may include:

- outgoing phonon phases,

- Kelvin-wave excitations,
- vortex displacement and writhe,
- density ripples,
- altered local reconnection possibilities,
- longer-range correlations in the surrounding medium.

The important point is not that every trace remains separately visible forever. The point is that the event is redistributed into the present state of the medium rather than physically unwritten.

### 4.1 The Wake Is Distributed

The wake of an event is generally not stored in one location. A reconnection, for example, may leave a local change in vortex geometry, a packet of outgoing phonons, a spectrum of Kelvin waves on the retracted filaments, and altered phase correlations in the surrounding medium. After further interactions these traces may become individually hard to identify, but their information has not thereby been gathered into the unique configuration needed to restore the pre-event state.

This distributed character is what makes wake entropy different from a memory metaphor. The medium does not merely "remember" in an observer-dependent sense. Its present physical state has been changed in ways that affect what subsequent local updates are possible.

### 4.2 Why Reconnection Matters But Does Not Exhaust the Story

Vortex reconnection remains an important process in SSV because it is one of the clearest mechanisms by which localized structure is converted into distributed wake structure. A reconnection event changes topology, emits phonons, populates Kelvin-wave sectors, and alters future reconnection possibilities. But the present paper does not identify entropy with a bare count of reconnections. Reconnection is one important wake-writing process inside a broader picture in which any event that leaves persistent structure in $\Psi$ contributes to the irreversible burden carried by the present state.

### 4.3 Local Cancellation Versus Global Unwriting

Local amplitudes can interfere destructively. Some observables can relax, thermalize, or become coarse-grained away. But this does not by itself restore the pre-event global state. To reverse an event physically would require the distributed wake to be gathered and re-focused into the unique incoming configuration that recreates the earlier state.

That is a nonlocal requirement. No local dynamical process in the medium is known to perform it.

## 5 Entropy as Wake-Complexity

This motivates a reframing of entropy in SSV. Entropy should not be introduced as a bare count of past events. That would risk tautology if monotonicity is built into the definition from the start. The stronger claim is:

> Entropy is the irreducible wake-complexity of the present state of $\Psi$.

## 5.1 Why This Is Not Tautological

The monotonicity claim is not:

> Entropy increases because we defined it to count only positive increments.

It is instead:

> Events write distributed structure into $\Psi$, and no physical process available to the medium globally unwrites that structure back into the exact pre-event state.

This is a physical claim about the availability of inverse processes, not merely a bookkeeping choice.

## 5.2 Relation to Boltzmann Entropy

The standard expression $S = k_B \ln \Omega$ remains useful; SSV changes the interpretation of the microstates being counted. A statistical form for wake entropy is

$$S_{\text{wake}}(t) = k_B \ln \Omega_{\text{wake}}[\Psi(t)], \tag{4}$$

where $\Omega_{\text{wake}}$ counts physically distinct present wake configurations compatible with the same coarse description, rather than an abstract set of hidden possibilities beneath a macrostate.

The phrase "the same coarse description" has to do real work. Concretely, "the same" means: equivalent under the smoothing of $\Psi$ to some scale $\ell$ and under restriction to a specified observable set $\mathcal{O}$. Two wake configurations are identified when their $\Psi$-fields, smoothed at scale $\ell$, agree on every observable in $\mathcal{O}$. Different choices of $(\ell, \mathcal{O})$ generate different $\Omega_{\text{wake}}$ and therefore different numerical entropies  a familiar feature of coarse-grained entropy, made explicit here so that $\Omega_{\text{wake}}$ is not a rename of $\Omega$ but a counting tied to the medium's resolvable wake channels.

This also explains why entropy can increase while the microscopic state remains definite. The medium has one actual state, but the macroscopic description loses access to the detailed distribution of wake structure. Thermodynamic entropy is then the coarse-grained shadow of a real structural burden in $\Psi$.

## 5.3 Candidate Wake Functional

Equivalently, one may write

$$S_{\text{wake}}(t) = k_B \, W[\Psi(t)], \tag{5}$$

where $W[\Psi(t)]$ measures the amount of present-state structure required to encode the surviving wake of prior events. A usable functional should respond to the parts of $\Psi$ that carry persistent wake structure. One possible schematic decomposition is

$$W[\Psi] = \alpha_\phi I_\phi + \alpha_\rho I_\rho + \alpha_{\mathcal{V}} I_{\mathcal{V}} + \alpha_{\mathcal{K}} I_{\mathcal{K}} + \alpha_{\mathcal{C}} I_{\mathcal{C}}, \tag{6}$$

where the $I$ terms measure phase disorder, density disorder, vortex tangle complexity, Kelvin-wave excitation, and correlation complexity respectively. The coefficients $\alpha_i$ set the relative thermodynamic weight of each channel.

Any acceptable $I_i$ must satisfy three requirements. It must be a functional of the present state, it must be insensitive to purely reversible relabellings, and it must increase when localized order is converted into distributed wake structure that cannot be locally unwritten.

This subsection states the decomposition as a checklist; Section 6 then carries it out, defining each $I_i$ as a concrete dimensionless functional, fixing each coefficient $\alpha_i$ by calibration to a known thermodynamic or counting limit, and reducing each channel to a recognizable entropy. The outcome is summarized in §6.6. The reader who wants the schematic picture first may treat the present equation as a checklist and read that section as its execution.

### 5.4 Fate of the Wake Channels

Although the $I_i$ are not yet canonical functionals, their physical content is concrete enough to classify by fate. Each channel of $W[\Psi]$ corresponds to wake content with a characteristic mode of propagation, decay, or persistence in the medium, and the irreversible burden carried by the present state is not distributed evenly across them. Table 1 summarizes the dominant fate of each channel.

| Channel | Dominant fate | Notes |
|---|---|---|
| $I_\phi$ (phase disorder) | radiated, then thermalized | Carried outward by phonons; mixed into phonon bath on scattering timescales. |
| $I_\rho$ (density disorder) | radiated, then thermalized | Sound-wave content; integral mass conservation gives a global constraint only. |
| $I_\mathcal{V}$ (vortex tangle) | conserved between events | Line length, writhe, and topology approximately conserved between reconnections; converted at events into $I_\mathcal{K}$ and $I_\phi$. |
| $I_\mathcal{K}$ (Kelvin-wave) | radiated along filaments | Kelvin cascade transports to small scales; ultimately decays into $I_\phi$ via Kelvin-phonon coupling at short wavelength. |
| $I_\mathcal{C}$ (correlation) | erased from observation | Long-range entanglement with the environment; structurally present but typically below resolved $(\ell, \mathcal{O})$. |

Table 1: Dominant fate of each wake channel under forward evolution in $\Psi$. "Radiated" means the content propagates outward but remains within $\Psi$; "thermalized" means it has mixed with bath modes such that the macroscopic description loses access to it; "erased from observation" means it persists in $\Psi$ but falls below the coarse-graining $(\ell, \mathcal{O})$ of any realistic measurement.

Three structural points follow. First, no channel is genuinely *lost* every fate column other than "conserved" transfers wake content to another channel or to a sector below resolution, never out of $\Psi$. Second, vortex tangle $I_\mathcal{V}$ is the only channel that carries wake across long timescales without converting; this is why reconnection acts as the rate-limiting event for irreversibility in vortex-dominated regimes (§4.2). Third, the classification is observer-relative only in the last column: the wake content tagged as "erased from observation" is real structure in $\Psi$ that is missed by a specific $(\ell, \mathcal{O})$, not absent from the medium. The first quantity of §5.7 retains it; the second loses it.

### 5.5 Extensivity, Coarse-Graining, and Present Structure

The wake formulation is not meant to deny the usefulness of ordinary thermodynamic entropy. It proposes a deeper physical interpretation for it: standard thermodynamic entropy is recovered at the macroscopic level by coarse-graining over the fine wake structure of the medium. On this view, extensivity and additivity are not primitive axioms, but emergent properties of how wake structure composes across weakly coupled subsystems.

### 5.6 The Initial-Condition Question

A monotonicity claim for $W[\Psi]$ has an immediate consequence: there must have been an earliest moment at which $W$ was minimal. This is the SSV analogue of the past hypothesis in standard cosmology the assumption that the universe began in a low-entropy state, as emphasized by Penrose's Weyl-curvature hypothesis [8] and developed in philosophical detail by Albert [7] and Price [6]. The present paper does not attempt to derive that condition. It treats it as a separate problem deferred to a later cosmological paper in the SSV program, where the initial condition on $\Psi$ should be addressed alongside the structure of the early medium. What the present paper does claim is weaker and self-contained: *given* a low-wake initial state of $\Psi$, the dynamics described in §4 and §8 cannot locally undo wake accumulation, and $W[\Psi(t)]$ therefore rises along physical histories. Whether the initial condition itself is natural, fine-tuned, or derivable from deeper structure is a question outside the scope of this paper.

### 5.7 Two Quantities, Not One: Where the Observer Enters

The preceding subsections actually describe two distinct quantities, and conflating them is the source of the standard worry about whether entropy is "in the world" or "in the head." The wake functional $W[\Psi]$ is a property of $\Psi$ alone: it is observer-independent, in the same sense that the density $\rho(\mathbf{x}, t)$ is observer-independent. The coarse-grained entropy that follows from $\Omega_{\text{wake}}$ depends on the choice of $(\ell, \mathcal{O})$, and is therefore observer-relative in the standard thermodynamic sense.

SSV takes both seriously. There is a structural burden carried by $\Psi$ regardless of description, and there is a coarse-grained shadow of that burden read off by an observer who resolves the medium only down to a particular scale and observable set. The two coincide in the limit where $(\ell, \mathcal{O})$ resolves every wake channel; they diverge wherever they do not. The monotonicity argument of §5.1 belongs to the first quantity. The recovery of ordinary thermodynamics in §5.5 belongs to the second.

### 5.8 Possible Empirical Signatures

The present paper does not advance a closed prediction, but the wake formulation suggests where wake entropy could in principle diverge from standard thermodynamic entropy. The most promising regimes are those where the medium's vortex and Kelvin-wave sectors carry a disproportionate share of $W[\Psi]$ relative to thermal degrees of freedom.

Three illustrative regimes are worth naming. (i) Decaying vortex tangles in superfluid $^4$He [17]: in the late-time decay of quantum turbulence, conventional thermodynamic entropy counts only the phonon bath, while $\alpha_{\mathcal{V}} I_{\mathcal{V}} + \alpha_{\mathcal{K}} I_{\mathcal{K}}$ would track the residual tangle and Kelvin-wave content that has not yet thermalized. A measured discrepancy between bulk thermal entropy and a geometry-aware estimator over the same volume would be evidence for the wake decomposition. (ii) BEC reconnection cascades [18]: the controlled reconnection of vortex filaments in Bose-Einstein condensates allows direct imaging of the topological event and its outgoing radiation, and is therefore a natural laboratory for comparing event-by-event wake deposition against post-reconnection thermal balance. (iii) Two coarse-grainings of the same state: any setting in which $(\ell, \mathcal{O})$ can be varied while the underlying $\Psi$ is held fixed should exhibit the predicted dependence of coarse-grained entropy on the resolved wake channels, in the manner described in §5.2.

None of these constitute a quantitative prediction in the present paper. They mark the regimes where the formalism developed here would most plausibly first be tested.

## 6 Channel-wise Reductions of the Wake Functional

This section carries out the reductions promised in §5.3. Each of the five channels of $W[\Psi]$ is given a concrete dimensionless definition, calibrated by matching to a known thermodynamic or counting limit, and reduced to a recognizable entropy form. The final subsection confirms that the channel set is complete and that all coefficients equal $k_B$.

## 6.1 Phonon-Bath Reduction

The preceding claim is concrete enough to demonstrate on a single channel. Consider a region of $\Psi$ supporting a thermal bath of Bogoliubov phonons at temperature $T$. The Madelung phase $\theta(\mathbf{x}, t)$ decomposes into mode amplitudes; each long-wavelength mode is a harmonic oscillator with frequency $\omega_k = c_s k$ and thermal occupation

$$n_k = \frac{1}{e^{\hbar\omega_k/k_B T} - 1}. \tag{7}$$

For the phonon channel, take $I_\phi$ to count the per-mode von Neumann entropy of the bulk acoustic excitations,

$$I_\phi \;=\; \sum_k \left[(1+n_k)\ln(1+n_k) - n_k \ln n_k\right]. \tag{8}$$

This is dimensionless, manifestly a functional of the present state of $\Psi$, and vanishes on the zero-temperature ground state ($n_k = 0$ for all $k$). Note that for Bogoliubov phonons, the phase and density quadratures are conjugate quadratures of the *same* oscillator, so a separate "density" channel $I_\rho$ would double-count on this sector; $I_\rho$ should therefore be reserved for non-phonon density structure (defects, solitons, condensate-density inhomogeneity) outside the bulk acoustic bath.

Setting $\alpha_\phi = k_B$ and applying the coarse-graining $(\ell, \mathcal{O})$ that resolves mode occupations but discards intra-mode phase information,

$$\alpha_\phi I_\phi \;=\; k_B \sum_k \left[(1+n_k)\ln(1+n_k) - n_k \ln n_k\right] \;=\; S_{\text{thermal}}, \tag{9}$$

which is the standard thermodynamic entropy of a Bose gas of phonons. The familiar $S \propto T^3$ scaling in the acoustic limit follows from $D(\omega) \propto \omega^2$ in three dimensions; no further input is required.

Two structural lessons. First, the calibration $\alpha_\phi = k_B$ is fixed by matching to a known thermodynamic entropy, illustrating in concrete terms the normalization route promised in §5.3. Second, the reduction is exact at the level of the phonon channel and serves as a template: an analogous construction in each remaining channel (with the appropriate basis for $I_\mathcal{K}$, $I_\mathcal{V}$, $I_\mathcal{C}$, and a non-phonon $I_\rho$) would reduce $W[\Psi]$ to the full thermodynamic entropy under the union of those coarse-grainings. The next subsections carry out the reduction on the remaining four channels.

## 6.2 Kelvin Sector

The Kelvin sector is genuinely independent of the bulk phonon bath: Kelvin waves are transverse bending excitations of quantized vortex filaments, supported on the one-dimensional locus of the vortex core rather than in the three-dimensional bulk. Their dispersion in the long-wavelength limit is [17]

$$\omega_\mathcal{K}(k) \;\approx\; \frac{\hbar}{2m}\, k^2\, \Lambda(k\xi), \qquad \Lambda(k\xi) \;=\; \ln(1/k\xi) + O(1), \tag{10}$$

with $m$ the boson mass and $\xi$ the healing length; the slow logarithm reflects the long-range interaction along the filament. In thermal equilibrium each Kelvin mode is again a harmonic oscillator with occupation

$$n_k^\mathcal{K} \;=\; \frac{1}{e^{\hbar\omega_\mathcal{K}(k)/k_B T} - 1}. \tag{11}$$

Take $I_\mathcal{K}$ to count the per-mode von Neumann entropy of the Kelvin sector,

$$I_\mathcal{K} \;=\; \sum_k \left[(1+n_k^\mathcal{K})\ln(1+n_k^\mathcal{K}) - n_k^\mathcal{K} \ln n_k^\mathcal{K}\right], \tag{12}$$

where the sum runs over Kelvin modes on the filament. Setting $\alpha_{\mathcal{K}} = k_B$ and applying the analogous coarse-graining $(\ell, \mathcal{O})$ that resolves Kelvin-mode occupations but discards intra-mode phases,

$$\alpha_{\mathcal{K}} I_{\mathcal{K}} \;=\; S_{\text{Kelvin}}^{\text{thermal}}, \tag{13}$$

the standard thermodynamic entropy of a Bose gas of Kelvin quanta on the filament.

Two remarks. First, although the formula for the per-mode entropy is identical to the phonon case, the underlying mode basis is different: bulk acoustic modes labeled by $\mathbf{k} \in \mathbb{R}^3$ for $I_\phi$ versus one-dimensional filament-bending modes for $I_{\mathcal{K}}$. The sum $\alpha_\phi I_\phi + \alpha_{\mathcal{K}} I_{\mathcal{K}}$ is therefore genuinely additive and does *not* double-count, in contrast to the conjugacy issue flagged in §6.1 for a putative $I_\rho$ on the same phonon modes. Second, because $\omega_{\mathcal{K}}(k) \propto k^2\Lambda$ rather than $c_s k$, the low-temperature scaling of $S_{\text{Kelvin}}^{\text{thermal}}$ differs from the acoustic $T^3$ law; the Kelvin channel therefore contributes a thermodynamically distinct dependence on $T$ at low temperature, which is in principle observable in geometries where the Kelvin sector dominates over bulk phonons (slender vortex filaments at low density).

### 6.3 Vortex-Tangle Sector

The vortex-tangle channel differs in kind from the two preceding examples. Phonons and Kelvin waves are harmonic excitations, and their entropy is the thermal entropy of a Bose gas of modes. A vortex tangle is not harmonic: its wake content is *configurational and topological* — the geometric arrangement of the filament network, its total quantized line length, and the knotting and linking of its components. The appropriate entropy is therefore a configurational count, and this channel turns out to be the most direct realization of the $S = k_B \ln \Omega_{\text{wake}}$ form of §5.2.

Discretize the medium at the healing length $\xi$, which is the physical core size of a quantized vortex and the natural short-distance cutoff: two filament segments approaching within $\sim \xi$ either reconnect or cease to be independent. A tangle of total line length $L$ then occupies $\ell = L/\xi$ filament segments on a lattice of spacing $\xi$. Because vortex cores cannot overlap, the filament network is a self-avoiding set of closed loops (together with any open arcs terminating on boundaries). The number of distinguishable tangle configurations at fixed total length scales, to leading order, as

$$\Omega_{\mathcal{V}} \;\sim\; \mu_{\text{eff}}^{L/\xi}, \tag{14}$$

where $\mu_{\text{eff}}$ is a dimensionless constant of order unity — the connective constant of the self-avoiding filament ensemble. Its precise value is regularization-dependent (for self-avoiding walks on a simple cubic lattice $\mu \approx 4.68$); what is robust is that $\ln \mu_{\text{eff}}$ is a positive number of order unity. Section 7.1 sharpens this: knotting and linking do not enlarge $\mu_{\text{eff}}$ but enter through a universal subleading exponent, so that $\mu_{\text{eff}} = \mu$ is exactly the polygon connective constant.

The tangle-channel functional is then

$$I_{\mathcal{V}} \;=\; \ln \Omega_{\mathcal{V}} \;=\; \frac{L}{\xi} \ln \mu_{\text{eff}} \;+\; (\text{subleading loop-closure and loop-number terms}). \tag{15}$$

This is dimensionless ($L/\xi$ is a pure ratio), is a functional of the present state of $\Psi$ — the core locus is the set where $\rho \to 0$ with nonzero phase winding — vanishes on the vortex-free ground state ($L = 0$), and is invariant under rigid relabellings of the medium.

Setting $\alpha_{\mathcal{V}} = k_B$, as for the previous two channels,

$$\alpha_{\mathcal{V}} I_{\mathcal{V}} \;=\; k_B \ln \Omega_{\mathcal{V}} \;=\; S_{\text{config}}, \tag{16}$$

a bona fide Boltzmann configurational entropy of the vortex tangle. No coarse-graining over a mode bath is required here: the tangle configurations *are* the microstates, and the vortex channel is therefore the cleanest instance of the interpretation advanced in §5.2, where $\Omega_{\text{wake}}$ counts physically distinct present configurations rather than abstract hidden possibilities beneath a macrostate.

One subtlety is essential and, properly understood, strengthens the framework: *the tangle functional $I_{\mathcal{V}}$ is not individually monotonic.* In freely decaying quantum turbulence the line-length density falls — in the Vinen regime as $L \propto t^{-1}$ [17] — so $I_{\mathcal{V}}$ *decreases* with time. This does not violate the monotonicity of $W[\Psi]$. Tangle decay proceeds by reconnection, and each reconnection converts organized line length into Kelvin-wave excitation and radiated phonons: wake flows out of $I_{\mathcal{V}}$ and into $I_{\mathcal{K}}$ and $I_{\phi}$. The total $W = \sum_i \alpha_i I_i$ does not decrease, because the conversion is dissipative — an organized, low-entropy channel feeding high-entropy thermalized channels. Only the sum is protected; the individual channels exchange wake. This is exactly the fate-table entry of §5.4, in which $I_{\mathcal{V}}$ carries wake across long timescales until reconnection converts it, and it is why reconnection is the rate-limiting event for irreversibility in vortex-dominated regimes.

Finally, $I_{\mathcal{V}}$ is directly computable. The total line length $L$ is a straightforward readout of any simulated state of $\Psi$: locate the core locus as the set of density zeros carrying nonzero phase winding, and measure its length. The knotted-filament states and continuation sweeps of Paper I [1] are therefore immediate test data for the vortex channel, and a measured $I_{\mathcal{V}}(t)$ along a reconnection cascade — paired with $I_{\mathcal{K}}(t)$ and $I_{\phi}(t)$ — would exhibit the inter-channel wake transfer just described as an explicit budget that is conserved in sum.

### 6.4 Correlation Channel

The four channels treated so far are all *local* in character: each counts entropy carried by degrees of freedom identified at a point or along a filament — bulk acoustic modes, Kelvin modes, vortex line. The correlation channel $I_{\mathcal{C}}$ is different in kind. It measures wake that resides in no single region but in the *joint statistics* of separated regions: the imprint left when one event correlates parts of the medium that a local description would treat as independent.

The natural measure is mutual information. Partition the coarse-grained medium at scale $\ell$ into regions, and for two regions $A$ and $B$ at separation $r$ define

$$I(A\!:\!B) \;=\; S(A) + S(B) - S(A \cup B), \tag{17}$$

with $S(\cdot)$ the entropy of the coarse-grained reduced state. A subtlety must be settled before this can serve as a channel of $W[\Psi]$. A state in local thermal equilibrium already carries correlations — the equilibrium correlations dictated by the same mode occupations already counted in $I_{\phi}$ and $I_{\mathcal{K}}$. Including those in $I_{\mathcal{C}}$ would double-count. The correlation channel must therefore count only the *connected, long-range* mutual information: the part that remains once the contributions already attributed to the local channels are removed, and that persists at separations $r$ much larger than the healing length $\xi$ and the thermal correlation length $\lambda_T$. Write

$$I_{\mathcal{C}} \;=\; \sum_{\text{region pairs}} I_{\text{conn}}(A\!:\!B), \qquad I_{\text{conn}}(A\!:\!B)\big|_{r \lesssim \xi,\, \lambda_T} \;\equiv\; 0, \tag{18}$$

the sum running over pairs in the partition family, with the short-range equilibrium part subtracted by construction.

So defined, $I_{\mathcal{C}}$ is dimensionless, is a functional of the present state of $\Psi$, and vanishes on any state of local thermal equilibrium. Setting $\alpha_{\mathcal{C}} = k_B$,

$$\alpha_{\mathcal{C}} I_{\mathcal{C}} \;=\; k_B \sum_{\text{region pairs}} I_{\text{conn}}(A\!:\!B), \tag{19}$$

a genuine entropy in thermodynamic units.

This channel carries the conceptual payoff of the whole decomposition. The first three worked examples each reduced a channel to a recognizable entropy — two Bose-gas entropies and a Boltzmann configurational entropy. The correlation channel does the opposite: it is the channel that *has no counterpart in ordinary local thermodynamics*, and that is exactly why it is the formal home of the wake idea. Standard thermodynamic entropy is recovered as the limit

$$I_{\mathcal{C}} \to 0 \quad \Longrightarrow \quad W[\Psi] \to \alpha_\phi I_\phi + \alpha_\rho I_\rho + \alpha_{\mathcal{V}} I_{\mathcal{V}} + \alpha_{\mathcal{K}} I_{\mathcal{K}}, \tag{20}$$

the sum of local channel entropies — the standard, extensive, locally-additive thermodynamic entropy of §5.5. A medium in local equilibrium has $I_{\mathcal{C}} = 0$ and obeys ordinary thermodynamics. A medium carrying wake has $I_{\mathcal{C}} > 0$: its separated regions share records of a common past. The correlation channel is the precise statement that the past survives as present structure — the structure being the connected correlations that local thermodynamics discards.

Two remarks. First, $I_{\mathcal{C}}$ is the most observer-relative channel: mutual information depends on the partition $(\ell, \mathcal{O})$, so $I_{\mathcal{C}}$ belongs squarely to the second of the two quantities distinguished in §5.7 — the coarse-grained shadow, not the partition-independent burden carried by $\Psi$ itself. Second, like $I_{\mathcal{V}}$, $I_{\mathcal{C}}$ is not individually monotonic: wake correlations spread as the medium evolves, raising the number of correlated region pairs while diluting the mutual information carried by each. Only the total $W$ is protected; the correlation channel exchanges wake with the others as records disperse.

## 6.5 Solitonic Density Channel

The final channel is the non-phonon density channel $I_\rho$. As noted in §6.1, density and phase are conjugate quadratures of the same Bogoliubov oscillator, so $I_\rho$ must *not* count small-amplitude acoustic structure — that is already in $I_\phi$. What it counts is the genuinely non-perturbative localized density structure of the medium: solitonic objects — dark and grey solitons, density notches, and other codimension-one density defects — that are neither sound nor vortex line. They are identified in $\Psi$ by a density depletion exceeding a fixed fraction of $\bar{\rho}$, localized over a width of order the healing length $\xi$, and carrying no net circulation (which would make them vortices, counted in $I_{\mathcal{V}}$).

The channel is configurational, and the counting follows the template of §6.3. Consider $N$ such solitonic objects in volume $V$. Each occupies a phase-space cell $v_0 = c_0\, \xi^d$, with $c_0$ a dimensionless constant of order unity and $d$ the spatial dimension, and carries an internal-parameter multiplicity $g_{\text{int}}$ — the number of distinguishable depth and velocity states of the soliton family. The number of distinguishable configurations of the dilute soliton gas is

$$\Omega_\rho \;=\; \frac{1}{N!}\left(\frac{V}{v_0}\right)^N g_{\text{int}}^N, \tag{21}$$

the $N!$ removing permutations of identical objects. With Stirling's approximation the channel functional is

$$I_\rho \;=\; \ln \Omega_\rho \;=\; N\left[\ln\!\left(\frac{V}{N v_0}\right) + 1 + \ln g_{\text{int}}\right]. \tag{22}$$

This is dimensionless, is a functional of the present state of $\Psi$ (soliton number and parameters are readouts of the density and phase fields), vanishes on the soliton-free ground state ($N = 0$), and is invariant under rigid relabellings.

Setting $\alpha_\rho = k_B$,

$$\alpha_\rho I_\rho \;=\; k_B \ln \Omega_\rho \;=\; S_{\text{soliton gas}}, \tag{23}$$

which is the configurational entropy of a dilute gas of localized objects — the Sackur–Tetrode configurational term, with $v_0$ in place of the thermal phase-space cell. The reduction to a recognizable thermodynamic entropy is therefore immediate, as for the vortex channel.

Two remarks complete the parallel with the earlier channels. First, $I_\rho$ does not double-count: solitonic objects are non-perturbative and codimension-one, distinct both from the linearized phonon bath ($I_\phi$) and from the codimension-two circulating defects of $I_{\mathcal{V}}$. Second, $I_\rho$ is not individually monotonic. In three dimensions dark solitons are unstable to the snake instability and decay into vortex rings — a wake transfer from $I_\rho$ into $I_\mathcal{V}$ — while soliton collisions radiate phonons into $I_\phi$. As before, $N$ can fall, $I_\rho$ with it, and only the total $W$ is protected.

### 6.6 The Channel Set Is Complete

With the density channel in hand, all five channels of the candidate functional (§5.3) have been given concrete dimensionless definitions and calibrated coefficients. A structural result emerges. Each $I_i$ was defined as a genuine entropy measured in nats — a Bose-gas entropy for $I_\phi$ and $I_\mathcal{K}$, a Boltzmann configurational entropy for $I_\mathcal{V}$ and $I_\rho$, a connected mutual information for $I_\mathcal{C}$. The coefficient that converts each to thermodynamic units is then the same in every case,

$$\alpha_\phi = \alpha_\rho = \alpha_\mathcal{V} = \alpha_\mathcal{K} = \alpha_\mathcal{C} = k_B, \tag{24}$$

so the wake entropy collapses to

$$S_{\text{wake}} = k_B\, W[\Psi] = k_B \sum_i I_i. \tag{25}$$

The relative-weight problem flagged in §5.3 therefore dissolves: once each channel is defined as a proper entropy, there are no free relative weights, and the only dimensionful constant in the construction is $k_B$. The content did not disappear — it moved, from "what are the relative weights" to "what is the correct entropy functional for each channel," which is where it belongs. What remains is finer, and is taken up in Section 7: the subleading terms in each functional, the status of the regularization constants ($\mu_{\text{eff}}$, $v_0$, $g_{\text{int}}$), and the reduction of the channel-wise coarse-grainings to a single renormalization flow.

## 7 Subleading Structure, a Unified Flow, and the Reconnection Quantum

The completeness result of §6.6 left a finer residue — the subleading terms of the configurational channels, the status of their regularization constants, and the reduction of the five channel-wise coarse-grainings to one flow — and the treatment of reconnection in §4.2 was deliberately left qualitative. This section closes both. The parts are not independent: §7.1 shows that the regularization constants of every channel play one and the same role, §7.2 shows that the five coarse-grainings are sectors of one renormalization flow whose initial data are exactly those constants, and §7.3 shows that reconnection is the discrete event that drives that flow in vortex-dominated regimes.

### 7.1 Subleading Terms and the Status of the Regularization Constants

Three channels were given only to leading order: the configurational counts $I_\mathcal{V}$ and $I_\rho$, and the connected-part subtraction in $I_\mathcal{C}$. Completing them shows that the leftover constants $\mu_{\text{eff}}$, $v_0$, and $g_{\text{int}}$ are not three separate loose ends but three instances of one structure.

*Vortex tangle.* The leading term $(L/\xi)\ln\mu_{\text{eff}}$ of §6.3 counts a self-avoiding loop ensemble. The asymptotic enumeration of a single self-avoiding polygon of $n = L/\xi$ segments is known [19],

$$p_n \;\sim\; A\,\mu^n\, n^\theta, \qquad \theta \;=\; \alpha - 3 \;=\; -(1+3\nu), \tag{26}$$

with $\mu$ the connective constant of the regularization, $A$ a non-universal amplitude, and $\nu \approx 0.587597$ the self-avoiding-walk correlation-length exponent [20]. The exponent $\theta \approx -2.76$ is universal — fixed by the hyperscaling relation $\alpha = 2 - d\nu$ in $d = 3$ — and hence regularization-independent. A tangle of $\mathcal{N}$ closed loops of lengths $\{L_i\}$ summing to $L$ then has

$$\Omega_{\mathcal{V}} \;=\; \frac{1}{\mathcal{N}!}\prod_{i=1}^{\mathcal{N}}\left[\frac{V}{\xi^3}\,A\,\mu^{\,L_i/\xi}\,(L_i/\xi)^{\theta}\right], \tag{27}$$

the factor $V/\xi^3$ placing each loop's centroid and the $\mathcal{N}!$ removing loop relabellings. With Stirling's approximation,

$$I_{\mathcal{V}} \;=\; \underbrace{\frac{L}{\xi}\ln\mu}_{\text{leading, extensive}} \;+\; \underbrace{\mathcal{N}\ln\frac{e\,A\,V}{\mathcal{N}\,\xi^3}}_{\text{loop-number}} \;+\; \underbrace{\theta\sum_{i=1}^{\mathcal{N}}\ln\frac{L_i}{\xi}}_{\text{loop-closure}}. \tag{28}$$

The leading term is unchanged. The loop-number term is itself a Sackur–Tetrode configurational entropy — now of the gas of loops rather than of the solitons of §6.5 — and the loop-closure term carries the universal exponent $\theta$. Knotting and linking, which §6.3 tentatively absorbed into an enlarged $\mu_{\text{eff}}$, are settled here in the opposite direction: knotted polygons are exponentially generic [21], so the *unrestricted* count already grows as $\mu^n$, and demanding a definite topology *lowers* the growth constant rather than raising it. Knotting does not enlarge the base; it shifts the subleading exponent, $\theta \to \theta + N_K$ with $N_K$ the number of prime factors of the knot type. Hence $\mu_{\text{eff}} = \mu$ exactly — the connective constant of the unrestricted self-avoiding-polygon ensemble, with knot diversity already counted in it.

*Soliton gas.* The dilute-gas count of §6.5 is the leading term of a virial expansion. Restoring the pairwise soliton interaction $U(r)$,

$$I_\rho \;=\; N\left[\ln\frac{V}{Nv_0} + 1 + \ln g_{\text{int}}\right] \;-\; \frac{N^2}{V}\,B_2(T) \;+\; O\!\left(\frac{N^3}{V^2}\right), \qquad B_2(T) = \tfrac{1}{2}\int d^d r\,\big(1 - e^{-U(r)/k_BT}\big). \tag{29}$$

For Gross–Pitaevskii dark solitons the interaction is exponentially screened, $U(r) \approx U_0\,e^{-2r/\xi}$ [25], so $B_2$ is finite and of order $\xi^d$, with only a weak logarithmic temperature dependence, $B_2 \approx (\xi^d/4)\ln(U_0/k_BT)$ for $k_BT \ll U_0$. The virial term is therefore computable and small in the dilute regime $N\xi^d/V \ll 1$ in which the channel was defined. The internal multiplicity is computable in the same way: it is the soliton internal partition function,

$$g_{\text{int}}(T) \;=\; \frac{1}{v_{\text{res}}}\int_{-c_s}^{c_s} dv\; e^{-E(v)/k_BT}, \qquad E(v) = E_0\big(1 - v^2/c_s^2\big)^{3/2}, \tag{30}$$

the integral running over the grey-soliton velocity family and $v_{\text{res}}$ being the velocity-resolution cell.

*Correlation channel.* The connected-part subtraction of §6.4 is made precise as a relative entropy. Let $\rho_{AB}$ be the coarse-grained joint state of two regions and $\sigma_{AB}$ the maximum-entropy state consistent with all data local to each region and with correlations out to the equilibrium scales $\xi$ and $\lambda_T$. Then

$$I_{\text{conn}}(A\!:\!B) \;=\; S(\rho_{AB}\,\|\,\sigma_{AB}) \;=\; \operatorname{Tr}\rho_{AB}\big(\ln\rho_{AB} - \ln\sigma_{AB}\big). \tag{31}$$

Relative entropy is non-negative and vanishes if and only if $\rho_{AB} = \sigma_{AB}$, i.e. exactly when the state carries no correlation beyond local equilibrium; for regions separated by more than $\xi$ and $\lambda_T$ the reference $\sigma_{AB}$ factorizes and $I_{\text{conn}}$ coincides with the mutual information of §6.4. The relative-entropy form is what makes "subtract the short-range equilibrium part" exact rather than schematic, and it guarantees $I_{\text{conn}} \geq 0$ with equality on local equilibrium. To Gaussian order it reduces to a sum over connected two-point functions, $I_{\text{conn}} \approx \frac{1}{2}\sum_{r \gtrsim \xi,\lambda_T} |G_c(r)|^2/(\chi_A\chi_B)$, with $\chi$ the local susceptibilities — the explicit long-range subleading structure of the channel.

*The status of $\mu$, $v_0$, $g_{int}$.* The three constants left undetermined are now seen to be the same kind of object. Each is a phase-space-cell measure: $\mu$ fixes the configurational volume per healing length of filament, $v_0 = c_0\,\xi^d$ the configurational volume per soliton, and $g_{\mathrm{int}}$, through $v_{\mathrm{res}}$, the cell of the soliton internal-parameter space. All three are non-universal, all three are fixed by the short-distance regularization at scale $\xi$, and all three enter $W[\Psi]$ only additively, as the zero-point of a channel's entropy. They therefore cancel in entropy differences at fixed defect content; and because the configurational entropy of a decaying defect is transferred intact to the channels that receive its wake — as the reconnection budget of §7.3 makes explicit — they cancel in the total rate $dW/dt$ as well. This is precisely the role of $h^3$ in the Sackur–Tetrode entropy: a regularization constant that sets the entropy zero, fixed externally — there by quantum mechanics, here by the Third-Law requirement that $W \to 0$ on the defect-free, zero-temperature ground state — and irrelevant to every entropy change. Computing $\mu$, $v_0$, and $g_{\mathrm{int}}$ "from the microscopic field equation" is therefore not a missing universal result but a category error: they are non-universal by construction. What *is* universal in the subleading structure — the closure exponent $\theta = -(1 + 3\nu)$, the virial coefficient $B_2(T)$, and the relative-entropy form of $I_{\mathrm{conn}}$ — has been given in closed form above.

### 7.2 The Five Coarse-Grainings as One Renormalization Flow

The construction so far attached a separate coarse-graining $(\ell, \mathcal{O})$ to each channel. That is an economy of presentation, not a fundamental feature, and this subsection removes it: the five coarse-grainings are five sectors of one renormalization flow on $\Psi$.

Define a single block transformation $\mathcal{R}_b$ that smooths the order parameter from resolution $\xi$ to resolution $b\xi$ — a real-space block average, or equivalently the integration of every mode with wavenumber between $\Lambda/b$ and $\Lambda = \xi^{-1}$. As a map on states of the medium, $\mathcal{R}_b$ is completely positive and trace-preserving: it discards information while preserving the structure of a probability assignment. Two consequences follow at once.

*Monotonicity becomes a theorem.* Under any completely positive trace-preserving map, the relative entropy to a fixed reference cannot increase [23]. Take the reference to be the local-equilibrium state $\sigma$ compatible with the present defect content; it is by construction a fixed point of the flow, $\mathcal{R}_b\,\sigma = \sigma$. Then $S(\mathcal{R}_b\,\rho\,\|\,\sigma)$ decreases monotonically along the flow: the state moves monotonically toward local equilibrium as it is coarse-grained. Equivalently, the entropy produced per step of the flow is non-negative,

$$\frac{dS}{d\ln b} \;\geq\; 0, \tag{32}$$

and the wake functional is the total entropy generated by running the single flow from the microscopic scale to the observation scale $\ell$,

$$W[\Psi] \;=\; \int_0^{\ln(\ell/\xi)} d(\ln b)\,\frac{dS}{d\ln b}. \tag{33}$$

The monotonicity of $W$, posited in §5.1 as a physical claim about the unavailability of inverse processes, is here recovered as the data-processing inequality for a coarse-graining map [24] — the same statement in information-theoretic dress.

*The five channels are the sectors of one generator.* The integrand $dS/d\ln b$ is a single quantity, but the modes it removes at scale $b$ fall into disjoint classes. Any fluctuation of $\Psi$ about a local condensate background is exactly one of the following, by the symmetry of the order parameter together with the homotopy classification of its defects [22]:

- a smooth $U(1)$ phase (Goldstone) fluctuation, with its conjugate density partner — the Bogoliubov sector — contributing $dS_\phi/d\ln b$;

- a codimension-two topological defect classified by $\pi_1$ — a vortex line — whose coarse-grained geometry contributes $dS_{\mathcal{V}}/d\ln b$;
- a mode bound to such a defect — a Kelvin wave on the filament — contributing $dS_{\mathcal{K}}/d\ln b$;
- a codimension-one non-perturbative density defect — a soliton — contributing $dS_\rho/d\ln b$;
- a connected correlation between regions, not reducible to local data, contributing $dS_{\mathcal{C}}/d\ln b$.

These five classes are mutually exclusive and jointly exhaustive — that is the content of the homotopy classification together with the Goldstone decomposition — so the generator splits with no remainder,

$$\frac{dS}{d\ln b} \;=\; \sum_{i\,\in\,\{\phi,\rho,\mathcal{V},\mathcal{K},\mathcal{C}\}} \frac{dS_i}{d\ln b}, \qquad I_i \;=\; \int_0^{\ln(\ell/\xi)} d(\ln b)\,\frac{dS_i}{d\ln b}. \tag{34}$$

The five channels are thus not five maps but one flow resolved into its irreducible sectors, and the completeness asserted in §6.6 is re-derived as the completeness of this classification. The common coefficient $\alpha_i = k_B$ follows in the same breath: each $I_i$ is the one entropy functional $S$ evaluated on a different sector of the one flow, so no sector can carry a different prefactor.

Two structural points close the picture. First, the regularization constants of §7.1 are the *initial data* of the flow at $b = 1$. The flow itself — its generator, its fixed point, its sector decomposition — is universal, while the value of $S$ at the bottom of the flow is the non-universal cell count. This is why $\mu$, $v_0$, and $g_{\text{int}}$ could be left unspecified without loss: they are boundary conditions on a universal flow, not parameters of it. Second, the flow has the expected fixed-point structure. As $b \to \infty$ the correlation sector is exhausted, $I_{\mathcal{C}} \to 0$, and the medium reaches local equilibrium — exactly the limit of §6.4 in which ordinary thermodynamics is recovered. The defect sectors $I_{\mathcal{V}}$ and $I_\rho$ do not flow to zero under coarse-graining alone: they are topologically protected, and can be discharged only by a defect-changing event. In vortex-dominated regimes that event is reconnection, to which we now turn.

### 7.3 The Reconnection Quantum

Reconnection was introduced in §4.2 as one wake-writing process among others, and deliberately not used as a primitive entropy counter. The channel decomposition now permits the quantitative statement that was postponed there: a reconnection is the discrete event that advances the flow of §7.2, and its wake can be priced exactly.

*Per-event budget.* A reconnection exchanges two filament strands and leaves two cusps that retract, removing a line length of order the inter-vortex spacing $\delta = \mathcal{L}^{-1/2}$, with $\mathcal{L}$ the line-length density. The energy released is the line energy of the removed segment,

$$\Delta E_{\text{rec}} \;\approx\; \epsilon\,\delta, \qquad \epsilon \;=\; \frac{\rho_s\,\kappa^2}{4\pi}\,\ln(\delta/\xi), \tag{35}$$

with $\epsilon$ the vortex line tension and $\kappa = h/m$ the circulation quantum. Resolved on the channels, the event debits the vortex sector and credits the Kelvin and phonon sectors,

$$\Delta I_{\mathcal{V}} \;=\; -\frac{\delta}{\xi}\,\ln\mu \;<\; 0, \qquad \Delta I_{\mathcal{K}} + \Delta I_\phi \;>\; \frac{\delta}{\xi}\,\ln\mu, \tag{36}$$

the configurational entropy $(\delta/\xi)\ln\mu$ of the destroyed segment passing intact into the microstates of the emitted burst, and the inequality holding because that burst also thermalizes the released energy. Once the burst thermalizes at the ambient temperature $T$, the net wake deposited by one reconnection takes the Clausius form

$$\Delta W_{\text{rec}} \;=\; \Delta I_{\mathcal{V}} + \Delta I_{\mathcal{K}} + \Delta I_\phi \;=\; \frac{\Delta E_{\text{rec}}}{k_B T} \;\approx\; \frac{\epsilon\,\delta}{k_B T}, \tag{37}$$

in which the cell constant $\mu$ has cancelled: the configurational entropy lost from $I_{\mathcal{V}}$ is exactly the configurational entropy gained by the burst, and only the thermalization term survives. A reconnection is, in this precise sense, a *quantum of irreversibility*: a discrete event depositing a definite, computable increment of wake.

*Rate and thermodynamic consistency.* The reconnection rate per unit volume is fixed by dimensional analysis from the only quantities available, the circulation $\kappa$ and the line density $\mathcal{L}$ [26],

$$\dot{N}_{\rm rec} \;=\; c_{\rm rec}\,\kappa\,\mathcal{L}^{5/2}, \tag{38}$$

with $c_{\rm rec}$ a dimensionless constant of order unity. The wake-production rate from reconnections is then, using $\delta = \mathcal{L}^{-1/2}$,

$$\left.\frac{dW}{dt}\right|_{\rm rec} \;=\; \dot{N}_{\rm rec}\,V\,\Delta W_{\rm rec} \;=\; c_{\rm rec}\,\frac{\kappa\,\epsilon}{k_B T}\,\mathcal{L}^2\,V. \tag{39}$$

This is a non-trivial consistency check. Freely decaying quantum turbulence obeys the Vinen law $d\mathcal{L}/dt = -\beta\,\kappa\,\mathcal{L}^2$ [17], so the line energy is dissipated at the rate $\epsilon\,|d\mathcal{L}/dt|\,V = \epsilon\beta\kappa\mathcal{L}^2 V$, and the associated thermodynamic entropy production is that power divided by $T$. The wake framework reproduces it exactly, and identifies the phenomenological Vinen coefficient with the reconnection count, $\beta = c_{\rm rec}$. The wake-production rate is therefore not an independent posit: it *is* the Vinen dissipation, $dW/dt = \epsilon\,|\dot{L}|/k_B T$, resolved into discrete events.

*Kinematic signature.* The per-event picture is testable below the level of bulk rates. The post-reconnection strand separation follows the universal law $\delta(t) \approx A_{\pm}\,|t - t_0|^{1/2}$, with distinct prefactors on approach ($A_-$) and recession ($A_+$) [18]. The asymmetry $\Delta A = A_+ - A_-$ is the kinematic marker of irreversibility — a time-symmetric reconnection would have $\Delta A = 0$ — and the energy radiated into the Kelvin and phonon sectors scales as $\Delta E_{\rm rec} \propto \Delta A^2$. A measured $\Delta A$ in a controlled Bose–Einstein-condensate reconnection therefore fixes $\Delta W_{\rm rec}$ for that event directly, turning the reconnection quantum from an estimate into a measured number.

# 8 Formal Time Reversal and Physical Reversal

A central distinction is needed between formal and physical reversal.

## 8.1 Formal Reversal

In many equations, one may perform the transformation

$$t \to -t \tag{40}$$

and obtain a formally related trajectory. This is a statement about the symmetry structure of the equations.

## 8.2 Physical Reversal

Physical reversal would require the medium itself to perform the inverse history: gather the distributed wake, reconstruct the exact incoming phases and correlations, and restore the prior global state. In the language of §3, it would require $\mathcal{R}[\Psi]$ to act with the sign and content of an inverse update rather than a forward one  not merely a sign change on an external time coordinate, but a physical operation the medium must actually carry out.

The existence of a formal reverse solution does not imply the existence of such an operation. Without this distinction, the phrase "time-reversal symmetry" invites a misleading inference: that because an equation admits a formally reversed trajectory, the physical world must be able to run backward in the same sense. Formal reversal is a property of a mathematical description; physical reversal is a claim about what operations the medium can actually perform.

## 8.3 Boundary Data and the Inverse Problem

The formal reverse of a dissipated event is not usually generated by a small local adjustment. It requires exquisitely tuned incoming boundary data across the region into which the wake has spread. In ordinary language, one would have to arrange the phonons, phases, correlations, and vortex perturbations so that they converge on the event site in precisely the configuration that reconstructs the former state.

SSV therefore locates irreversibility in the gap between a formal inverse trajectory and an available physical operation. The reverse trajectory may exist in the mathematical solution space. The present medium does not thereby contain a machine that prepares its boundary conditions.

This is also where the standard Loschmidt [9] and Poincaré [10] objections are absorbed. Loschmidt's paradox notes that microreversible dynamics admit a formally reversed trajectory for every forward one; SSV does not deny that, but holds (with Price [6]) that the reverse trajectory is realized only when the required incoming boundary data are actually supplied which the medium does not do spontaneously. Poincaré recurrence guarantees return to an arbitrarily close state in a bounded system with discrete spectrum on cosmologically long timescales, but $\Psi$ in SSV is effectively unbounded and supports continuous-spectrum excitations (outgoing phonons, radiated Kelvin content, long-range correlations). The wake propagates into channels with no finite recurrence time on physically relevant scales. Both objections are about what the mathematics permits in the limit; the SSV claim is about what the medium delivers in finite time.

## 8.4 Classical Time Reversal Versus Quantum Unitarity

The phrase "time reversal" covers two different mathematical structures, and the wake argument is closer to one than the other. Classical time-reversal symmetry takes a trajectory $\{q(t), p(t)\}$ to $\{q(-t), -p(-t)\}$ and is the symmetry Loschmidt invoked. Quantum unitary evolution is more delicate: a closed system's state evolves under $U(t) = e^{-iHt/\hbar}$, and $U^\dagger(t)$ is the formal inverse backward evolution exists in principle for any pure state.

The wake argument applies to both, but for related-yet-distinct reasons. Against classical reversal it is the Loschmidt response just given. Against quantum reversal it is closer in spirit to einselection and decoherence [11, 12]: the wake of an event in $\Psi$ is information entangled with environmental degrees of freedom across a vast region of the medium, and the local subsystem no longer admits a pure-state description compatible with the pre-event state. SSV does not deny global unitarity; it claims that the globally unitary operation needed to physically reverse an event is not available to any local agent in $\Psi$, because the requisite environmental boundary conditions cannot be locally prepared. The distinction between formal and physical reversal is, in the quantum case, the distinction between the existence of $U^\dagger$ as an operator and the existence of an apparatus that implements it on a wake-dispersed state.

## 8.5 Application: Closed Timelike Curves and Quantum CTCs

The wake argument has direct consequences for physical proposals that invoke time travel as a literal possibility. Two cases are worth naming.

*Classical closed timelike curves.* General relativity admits solutions (Gödel [13], Kerr-interior, Tipler cylinder, traversable wormholes with exotic matter [14]) containing closed timelike curves. The usual interpretation is that a worldline could in principle return to its own past. In SSV this reading is incoherent for the reason developed in §3: there is no past region to return to. A closed timelike curve in the metric, if such a configuration of $\Psi$ were physically realizable at all, would have to be reinterpreted as a closed structure in the present state of the medium a peculiar geometry, not an access route to an earlier moment. The metric solution may exist; what does not exist is a destination at the other end of the loop.

*Deutsch's quantum CTCs.* Deutsch's D-CTC proposal [15] restores self-consistency by demanding a fixed-point density matrix on the chronology-violating region: $\rho_{\text{CTC}} = \text{Tr}_A[U(\rho_A \otimes \rho_{\text{CTC}})U^\dagger]$. The mathematical content is a fixed point of a quantum channel; it is closed under any unitary $U$ and famously implies computational consequences (efficient solution of NP-hard problems, distinguishability of non-orthogonal states; see also the postselected variant of Lloyd *et al.* [16]). The SSV response is structural rather than computational: the D-CTC prescription quietly assumes that the fixed-point state is *prepared* on the chronology-violating region, but preparation is itself a wake-writing operation in $\Psi$. There is no agent in the medium whose preparing the fixed-point state does not itself write distributed wake whose pre-state cannot be reassembled. Deutsch's formalism solves the consistency problem at the level of operators while ignoring the ontological problem at the level of $\Psi$: the wake of preparing the loop is already elsewhere by the time the loop is supposed to close on itself.

Both cases share the same structure. They treat a formal solution (a CTC metric, a fixed-point density matrix) as if the formal existence of the solution sufficed to make the physical operation available. SSV holds that this is the same category error as before, dressed in heavier mathematics.

## 9 Discussion and Open Problems

The paper makes a narrow ontological claim. The wake functional $W[\Psi]$ has been carried from a schematic checklist to a closed construction: all five channels are defined as concrete dimensionless functionals and calibrated to the single coefficient $k_B$ (Section 6); their subleading structure is given in closed form, the five channel-wise coarse-grainings are unified as one renormalization flow, and reconnection is priced as a discrete quantum of wake (Section 7). The three technical tasks left open at §6.6 are thereby resolved at the level of structure. What remains open is of a different and more limited kind:

1. evaluate the non-universal cell constants — $\mu$ for the vortex channel, $v_0$ and $g_{\text{int}}$ for the soliton channel, $c_{\text{rec}}$ for the reconnection rate — numerically, by direct enumeration on the Gross–Pitaevskii ground state. This is a computation, not a conceptual gap: §7.1 has shown that these constants set only the zero of each channel's entropy and cancel in every entropy difference and rate,
2. connect this time framework to gravity and black holes in later papers without overloading the present one, and in particular supply the cosmological initial condition on $\Psi$ left open at §5.6.

The gain from the present framing is conceptual clarity. If time is local change, and the past survives only as present wake, then irreversibility is not an added rule imposed on a reversible block universe. It is the physical consequence of how events are written into a real medium.

## 10 Conclusion

The payoff of the present framing is that irreversibility stops being an extra rule layered on top of a reversible block universe. If time is local change in $\Psi$ and the past survives only as wake structure in the present state, then the asymmetry of experience and the asymmetry of equations are no longer in tension: they describe different objects. Equations describe trajectory classes that admit formal reversal; the medium carries out forward updates whose distributed wake is not locally invertible. Entropy, on this account, is what that non-invertibility looks like when it is read off the present state of $\Psi$.

What later papers inherit from this one is therefore a constraint, not a theorem. Any SSV-derived account of gravitation, cosmology, or black-hole structure must be compatible with the claim that the past is not a place and any candidate entropy law for $\Psi$ must ultimately reduce to a wake-functional of the kind constructed in Sections 5–7. That functional has here been carried to closed form, channel by channel; what is left to subsequent papers is its coupling to gravitation and the cosmological initial condition on $\Psi$.

## References


[1] S. Norland, *Saturated Superfluid Vacuum I: Topological Defects in a Saturated Superfluid Vacuum — The Geometric Origin of Matter and the $\alpha$-Harmonic Mass Ladder*, 10.5281/zenodo.19651986, 2026.

[2] S. Norland, *Saturated Superfluid Vacuum II*, SSV program, 2026.

[3] J. Barbour, *The End of Time: The Next Revolution in Physics*, Oxford University Press, 1999.

[4] C. Rovelli, "Statistical mechanics of gravity and the thermodynamical origin of time," *Class. Quantum Grav.* **10**, 1549–1566 (1993).

[5] C. Rovelli, *The Order of Time*, Riverhead Books, 2018.

[6] H. Price, *Time's Arrow and Archimedes' Point: New Directions for the Physics of Time*, Oxford University Press, 1996.

[7] D. Z. Albert, *Time and Chance*, Harvard University Press, 2000.

[8] R. Penrose, "Singularities and time-asymmetry," in *General Relativity: An Einstein Centenary Survey*, eds. S. W. Hawking and W. Israel, Cambridge University Press, 1979, pp. 581–638.

[9] J. Loschmidt, "Über den Zustand des Wärmegleichgewichtes eines Systems von Körpern mit Rücksicht auf die Schwerkraft," *Sitzungsber. Akad. Wiss. Wien* **73**, 128–142 (1876).

[10] H. Poincaré, "Sur le problème des trois corps et les équations de la dynamique," *Acta Math.* **13**, 1–270 (1890).

[11] W. H. Zurek, "Decoherence, einselection, and the quantum origins of the classical," *Rev. Mod. Phys.* **75**, 715–775 (2003).

[12] E. Joos, H. D. Zeh, C. Kiefer, D. Giulini, J. Kupsch, and I.-O. Stamatescu, *Decoherence and the Appearance of a Classical World in Quantum Theory*, 2nd ed., Springer, 2003.

[13] K. Gödel, "An example of a new type of cosmological solutions of Einstein's field equations of gravitation," *Rev. Mod. Phys.* **21**, 447–450 (1949).

[14] M. S. Morris and K. S. Thorne, "Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity," *Am. J. Phys.* **56**, 395–412 (1988).

[15] D. Deutsch, "Quantum mechanics near closed timelike lines," *Phys. Rev. D* **44**, 3197–3217 (1991).

[16] S. Lloyd, L. Maccone, R. Garcia-Patron, V. Giovannetti, Y. Shikano, S. Pirandola, L. A. Rozema, A. Darabi, Y. Soudagar, L. K. Shalm, and A. M. Steinberg, "Closed timelike curves via postselection: Theory and experimental test of consistency," *Phys. Rev. Lett.* **106**, 040403 (2011).

[17] W. F. Vinen and J. J. Niemela, "Quantum turbulence," *J. Low Temp. Phys.* **128**, 167–231 (2002).

[18] A. Villois, D. Proment, and G. Krstulovic, "Universal and nonuniversal aspects of vortex reconnections in superfluids," *Phys. Rev. Fluids* **2**, 044701 (2017).

[19] N. Madras and G. Slade, *The Self-Avoiding Walk*, Birkhäuser, Boston, 1993.

[20] N. Clisby, "Accurate estimate of the critical exponent $\nu$ for self-avoiding walks via a fast implementation of the pivot algorithm," *Phys. Rev. Lett.* **104**, 055702 (2010).

[21] D. W. Sumners and S. G. Whittington, "Knots in self-avoiding walks," *J. Phys. A: Math. Gen.* **21**, 1689–1694 (1988).

[22] N. D. Mermin, "The topological theory of defects in ordered media," *Rev. Mod. Phys.* **51**, 591–648 (1979).

[23] G. Lindblad, "Completely positive maps and entropy inequalities," *Commun. Math. Phys.* **40**, 147–151 (1975).

[24] V. Vedral, "The role of relative entropy in quantum information theory," *Rev. Mod. Phys.* **74**, 197–234 (2002).

[25] D. J. Frantzeskakis, "Dark solitons in atomic Bose–Einstein condensates: from theory to experiments," *J. Phys. A: Math. Theor.* **43**, 213001 (2010).

[26] C. F. Barenghi, L. Skrbek, and K. R. Sreenivasan, "Introduction to quantum turbulence," *Proc. Natl. Acad. Sci. USA* **111**, 4647–4652 (2014).